

POOLED leave-one-node-out  $R^2$  — both directions (top: A→B, bottom: B→A; ridge top-6 PC,  $\alpha=1000$ ; best-cell node-shuffle null, 200 perms)

best-cell null  $-0.19 \pm 0.14$ ,  $p=0.005$

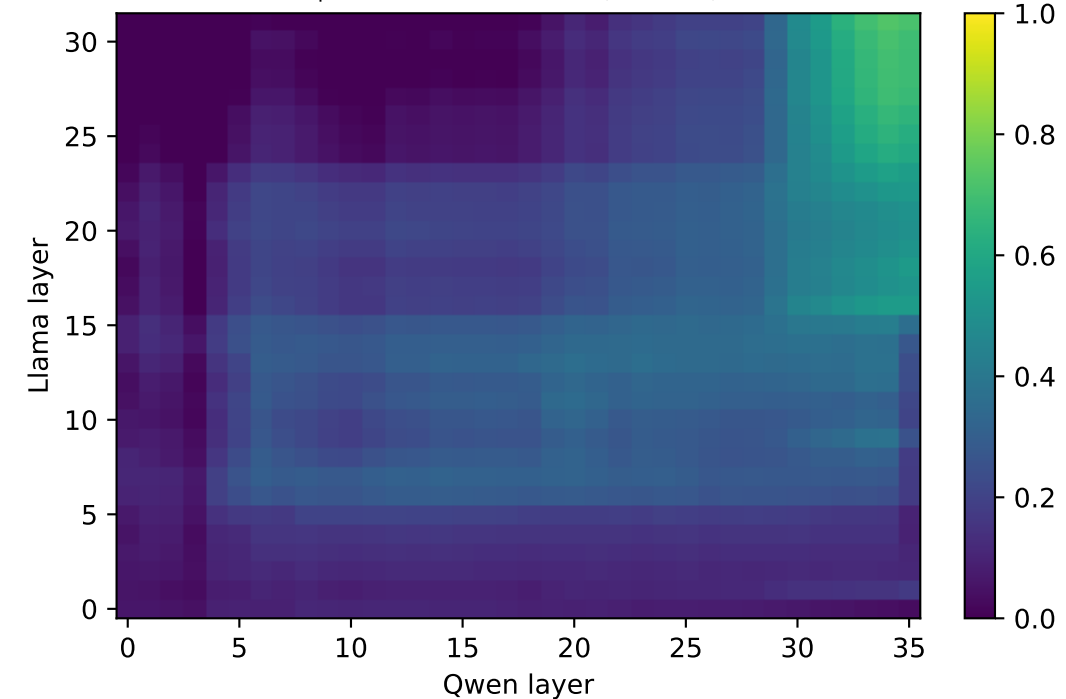
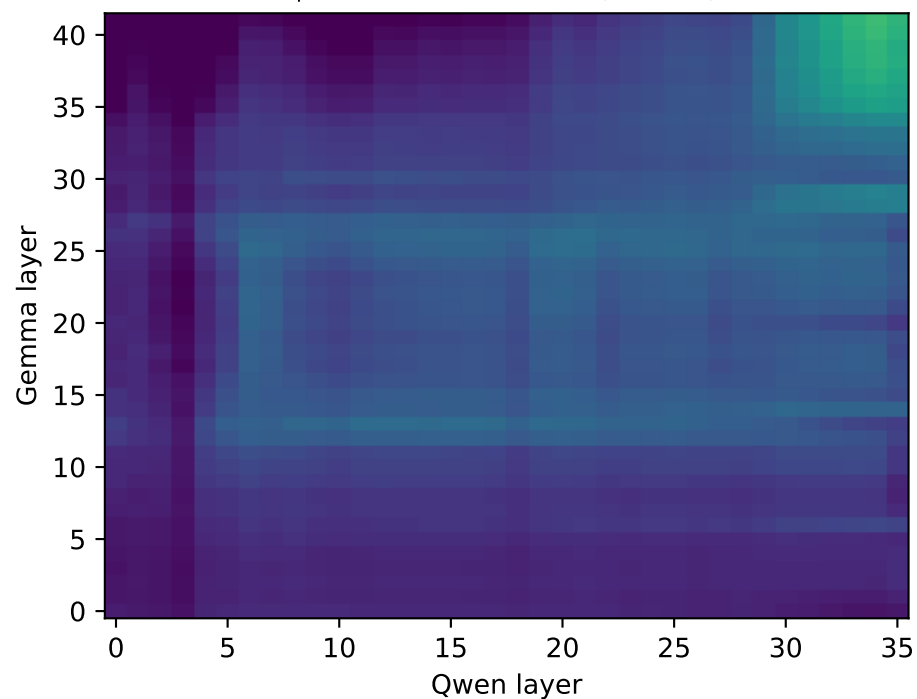
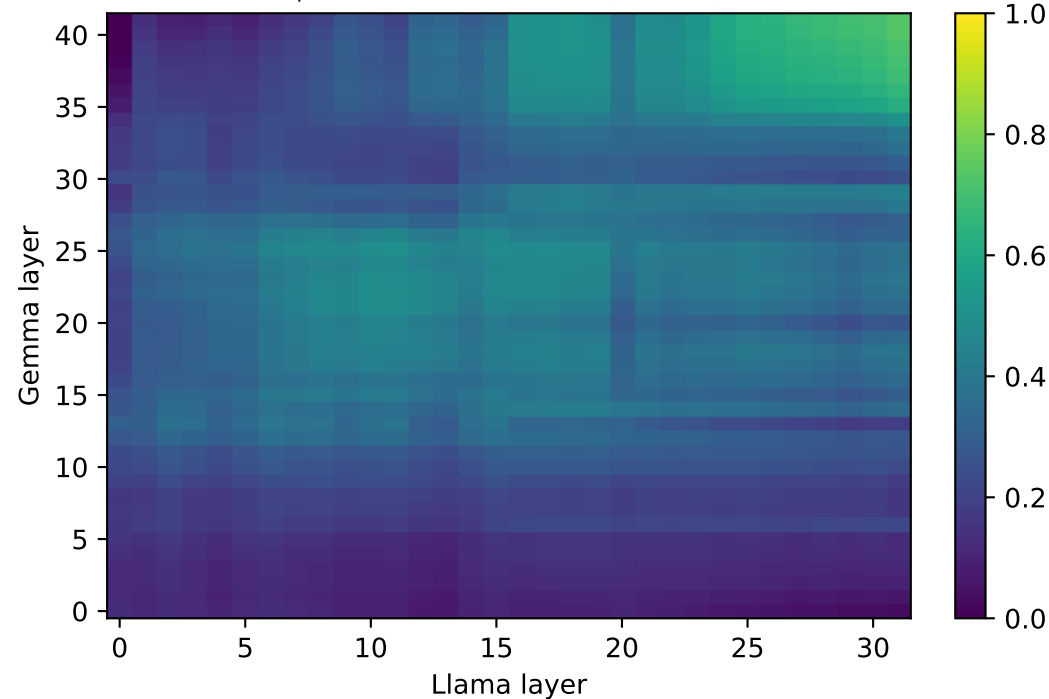
best-cell null  $-0.20 \pm 0.14$ ,  $p=0.005$

best-cell null  $-0.27 \pm 0.17$ ,  $p=0.005$

predict Llama from Gemma (max 0.73)

predict Qwen from Gemma (max 0.68)

predict Qwen from Llama (max 0.73)



best-cell null  $-0.27 \pm 0.14$ ,  $p=0.005$

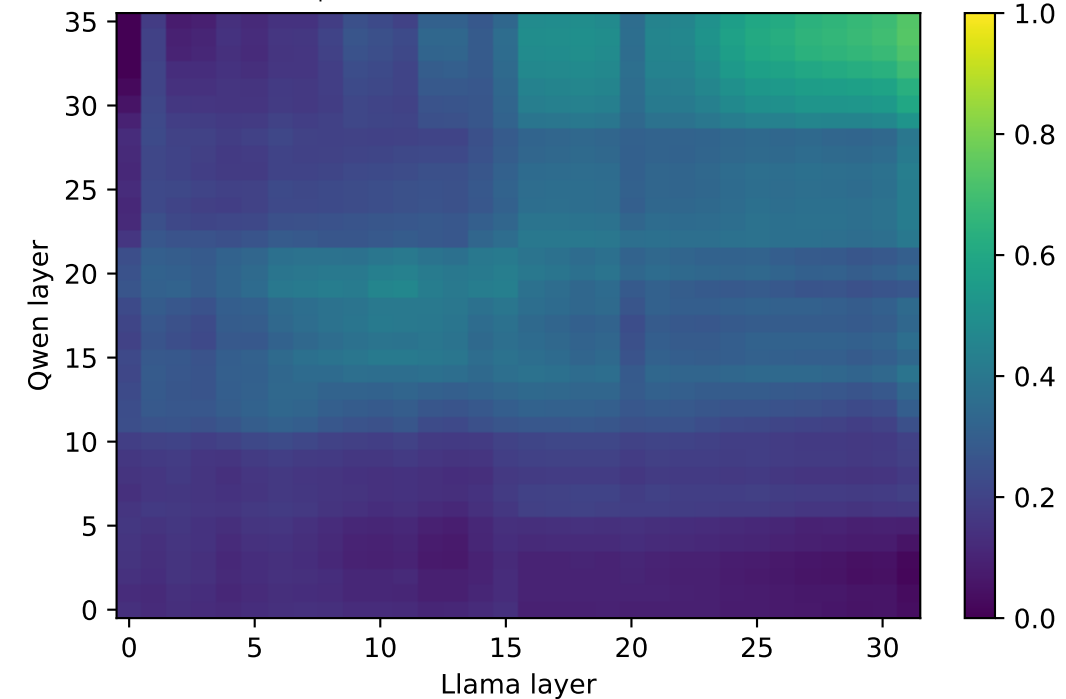
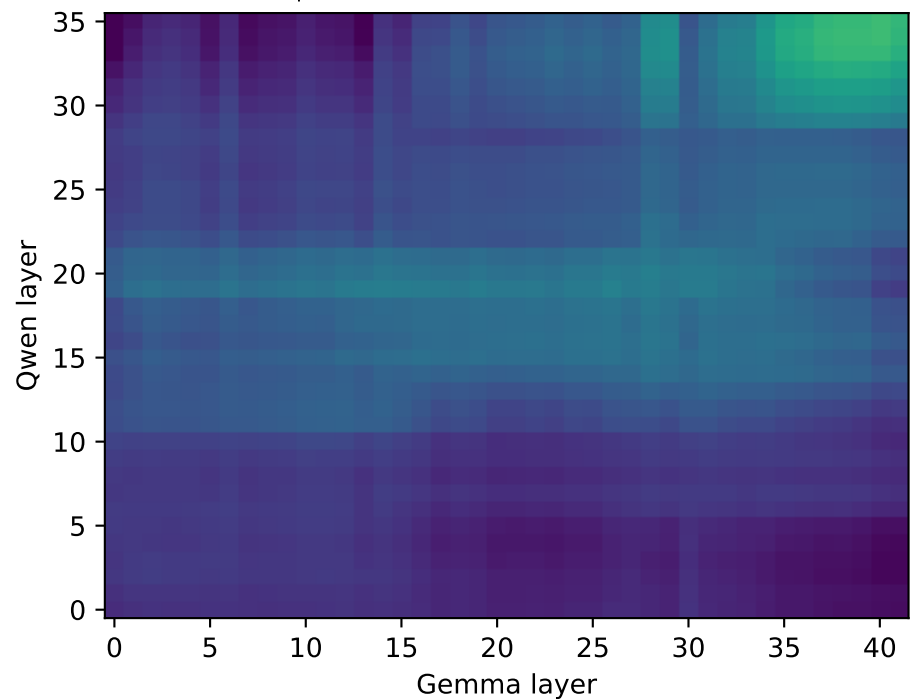
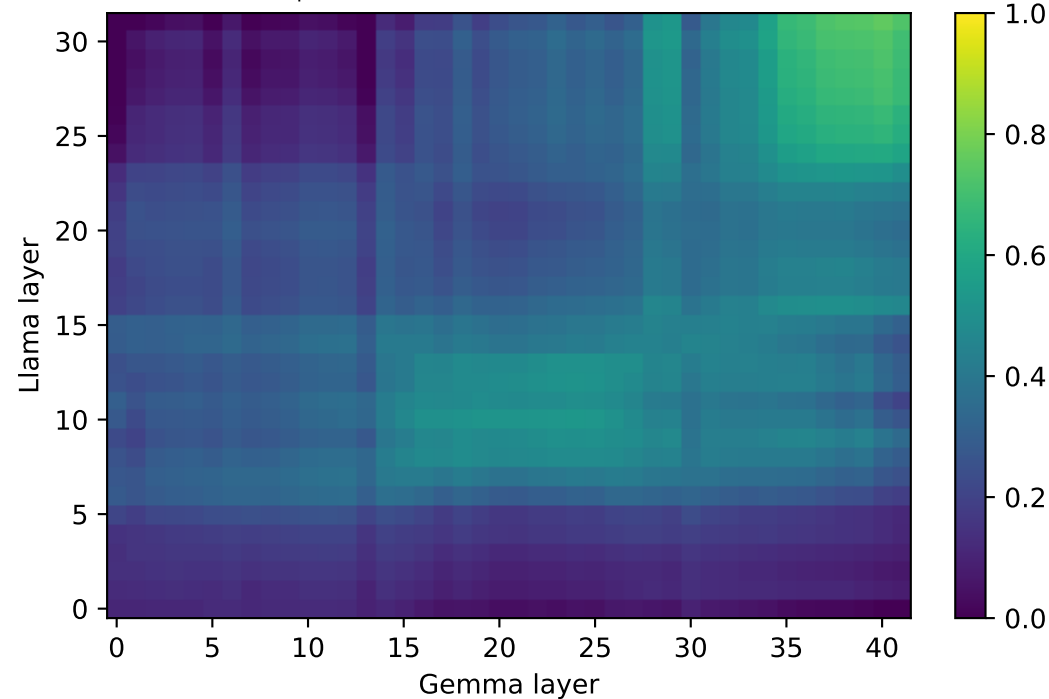
best-cell null  $-0.22 \pm 0.11$ ,  $p=0.005$

best-cell null  $-0.19 \pm 0.15$ ,  $p=0.005$

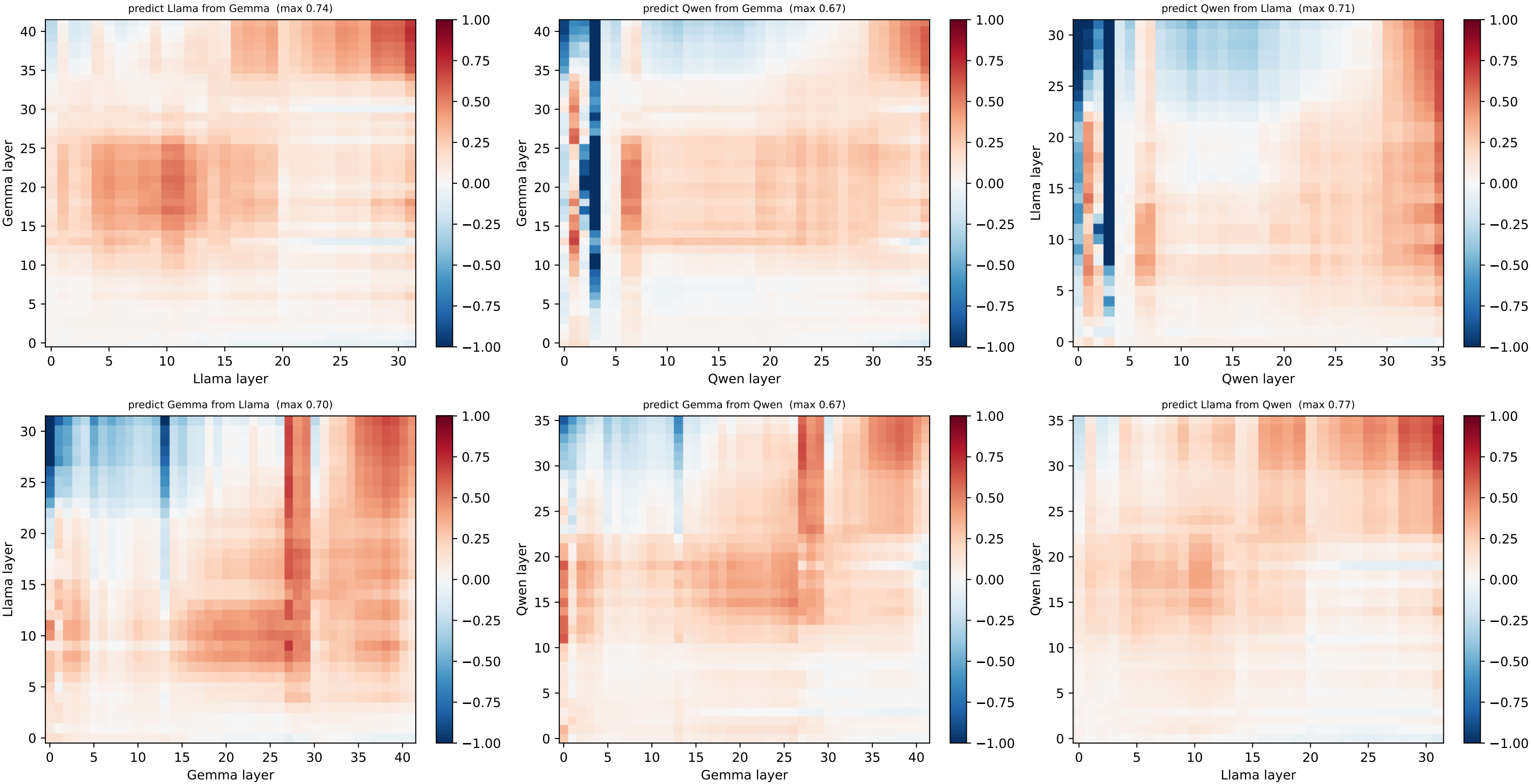
predict Gemma from Llama (max 0.76)

predict Gemma from Qwen (max 0.69)

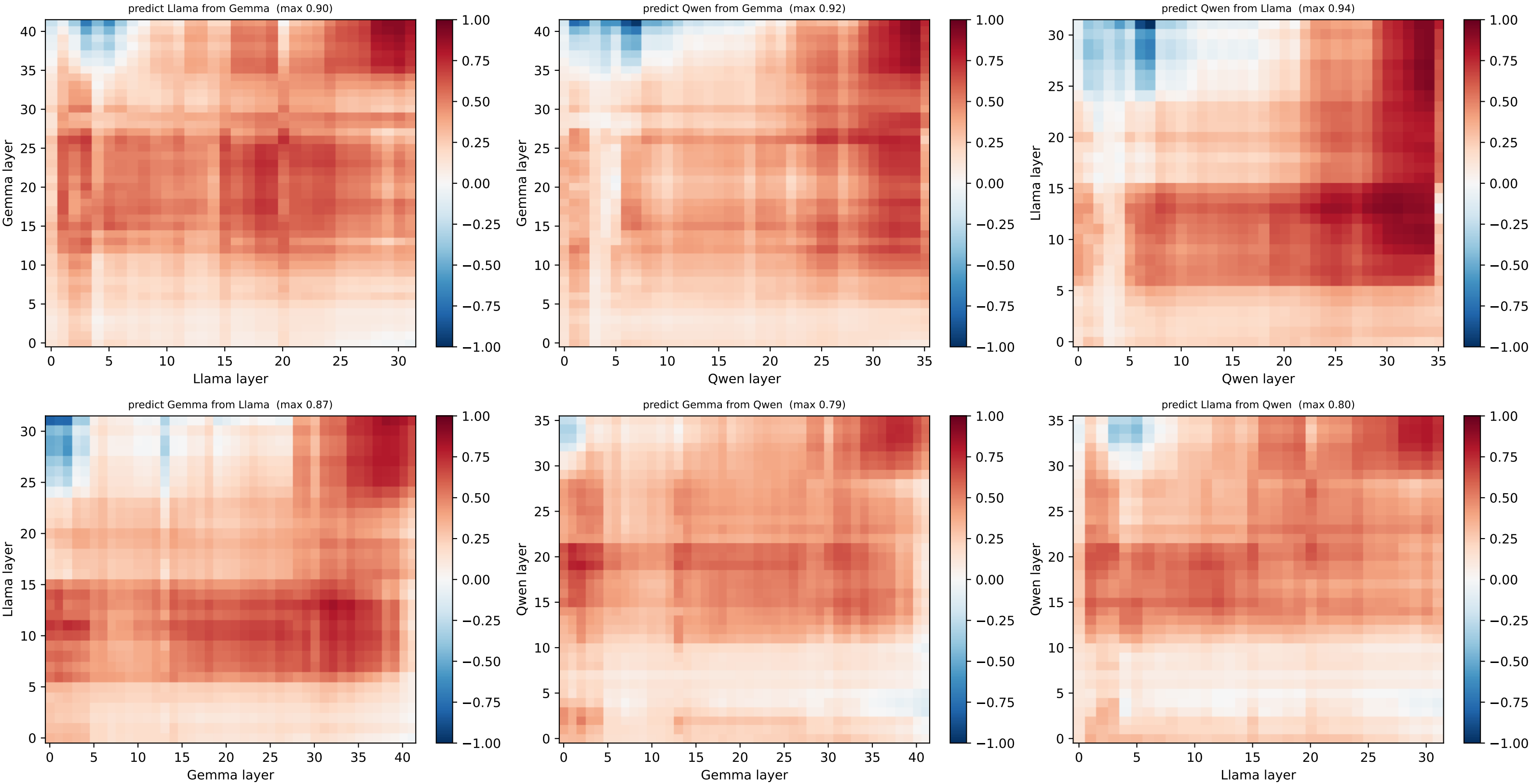
predict Llama from Qwen (max 0.74)



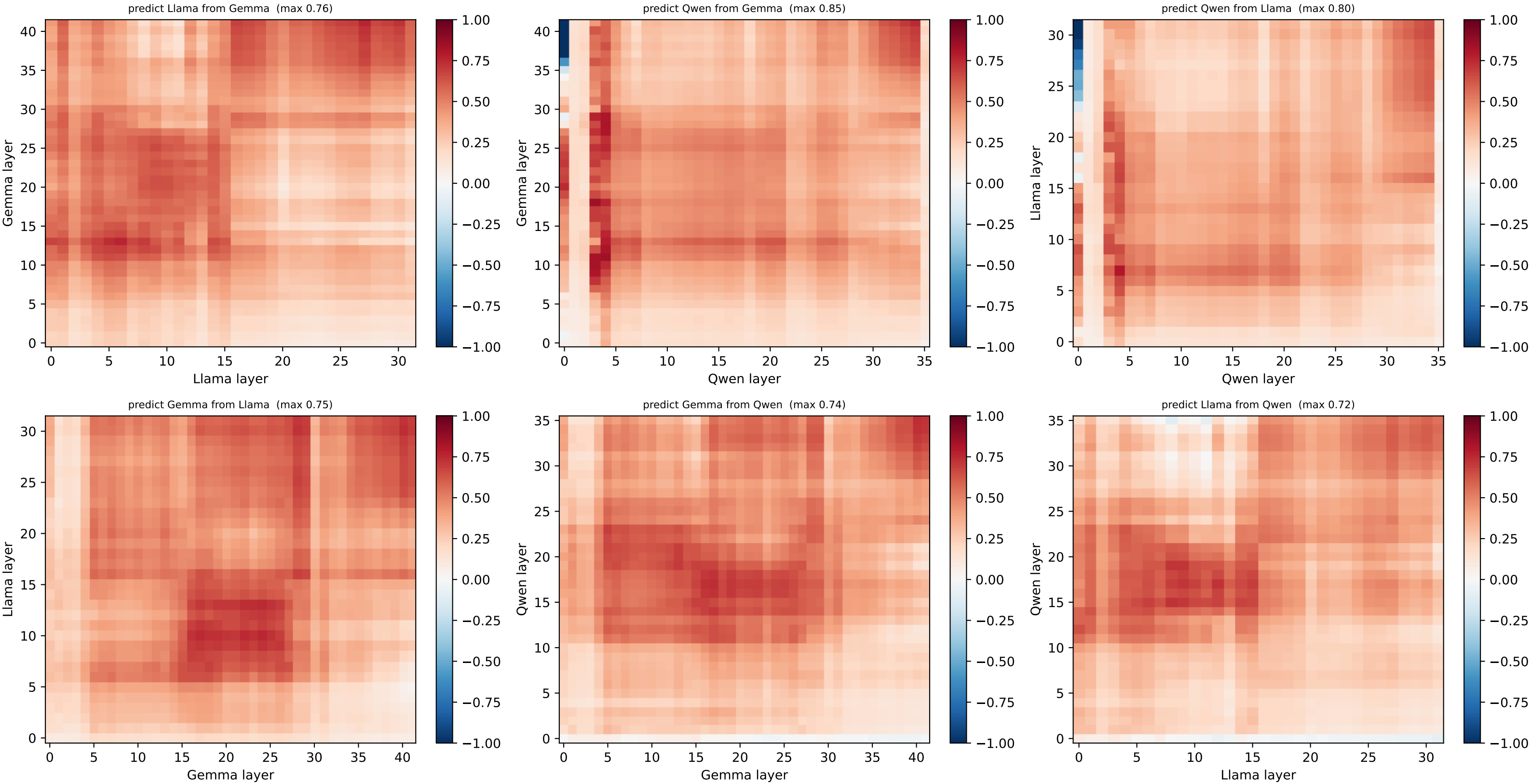
[grid] HELD-OUT node 0 @ (0.0, 0.0) — per-node  $R^2$  layer×layer, both directions (red=well predicted, blue=worse than centroid)



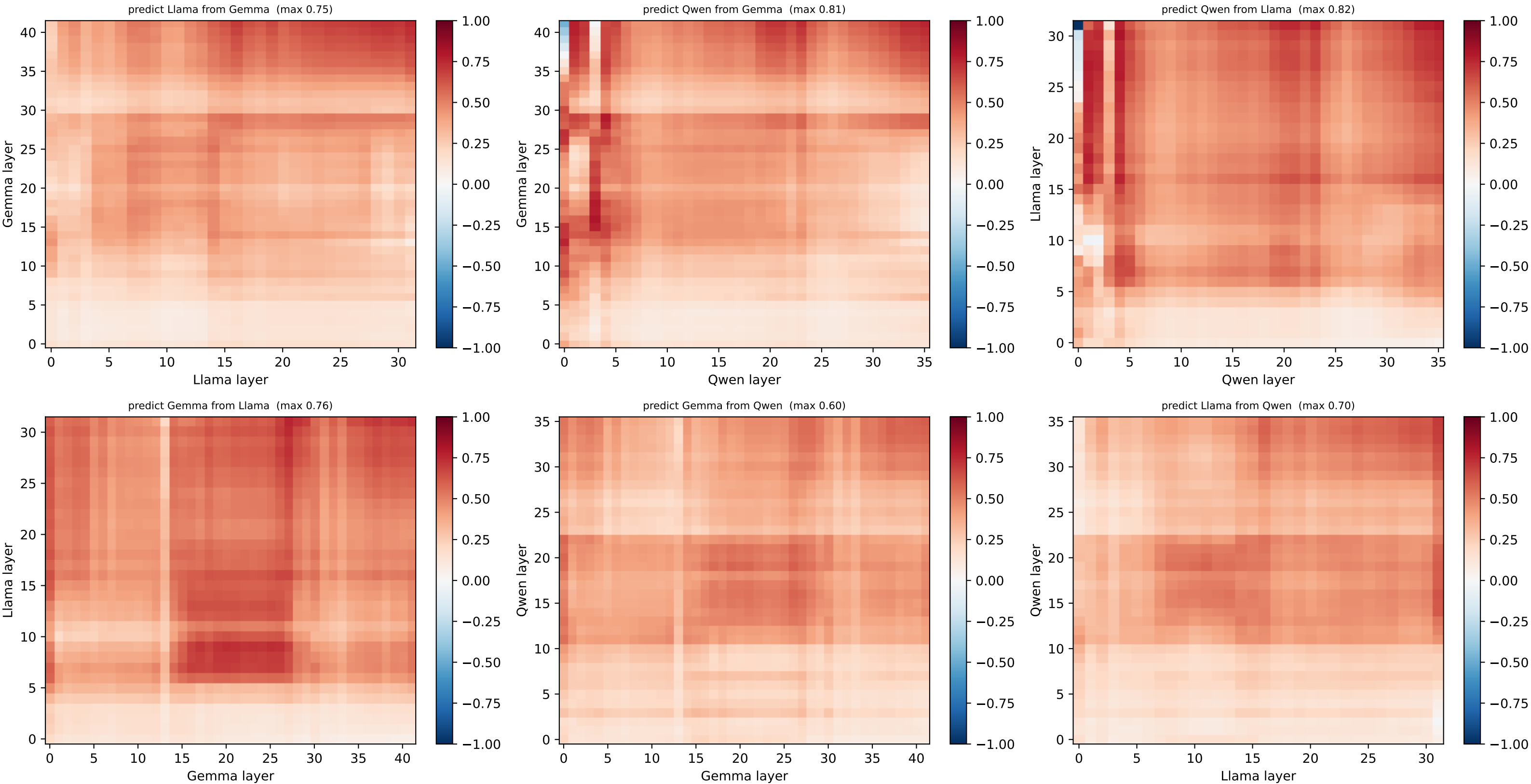
[grid] HELD-OUT node 1 @ (0.0, 1.0) — per-node  $R^2$  layer×layer, both directions (red=well predicted, blue=worse than centroid)



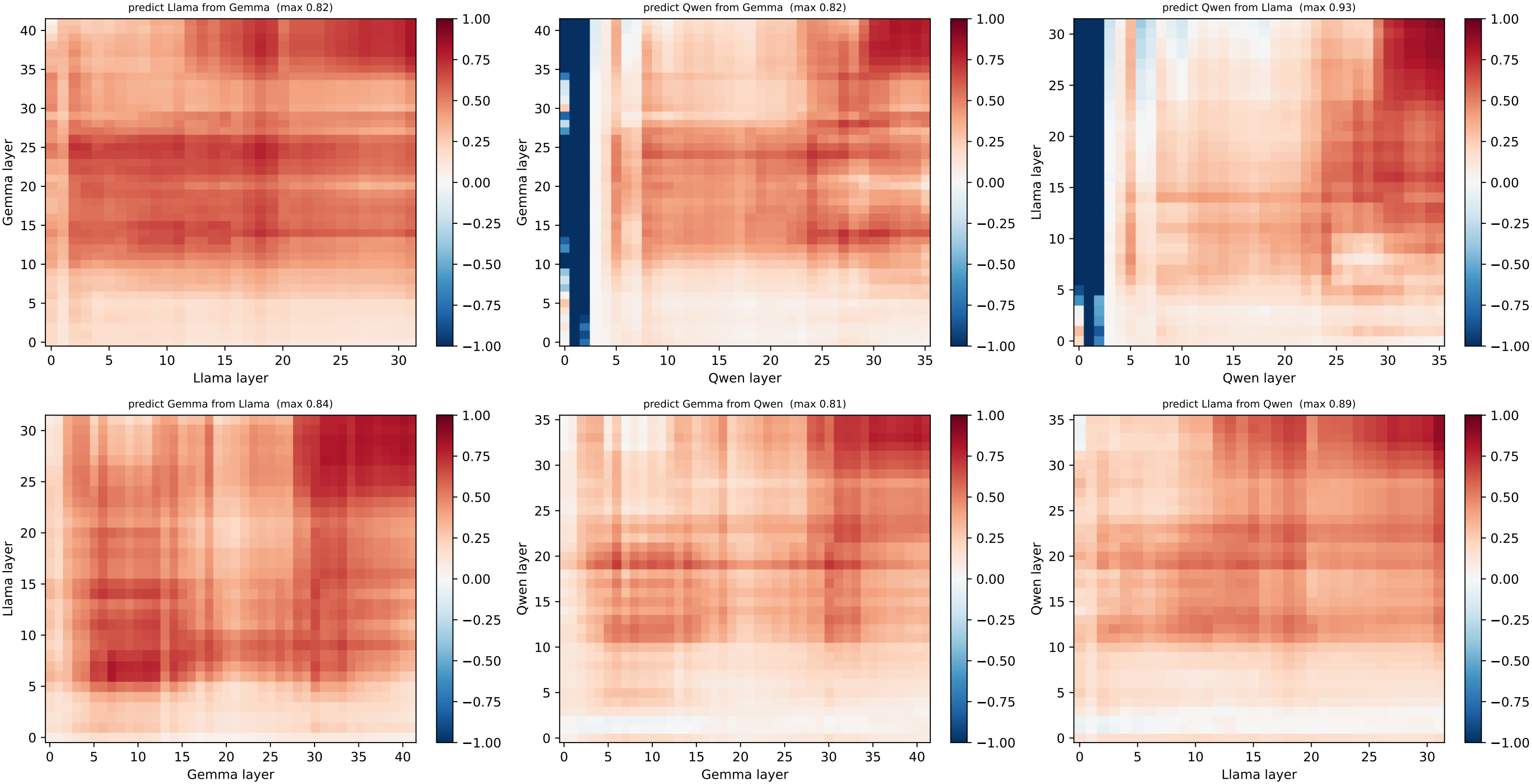
[grid] HELD-OUT node 2 @ (0.0, 2.0) — per-node  $R^2$  layer×layer, both directions (red=well predicted, blue=worse than centroid)



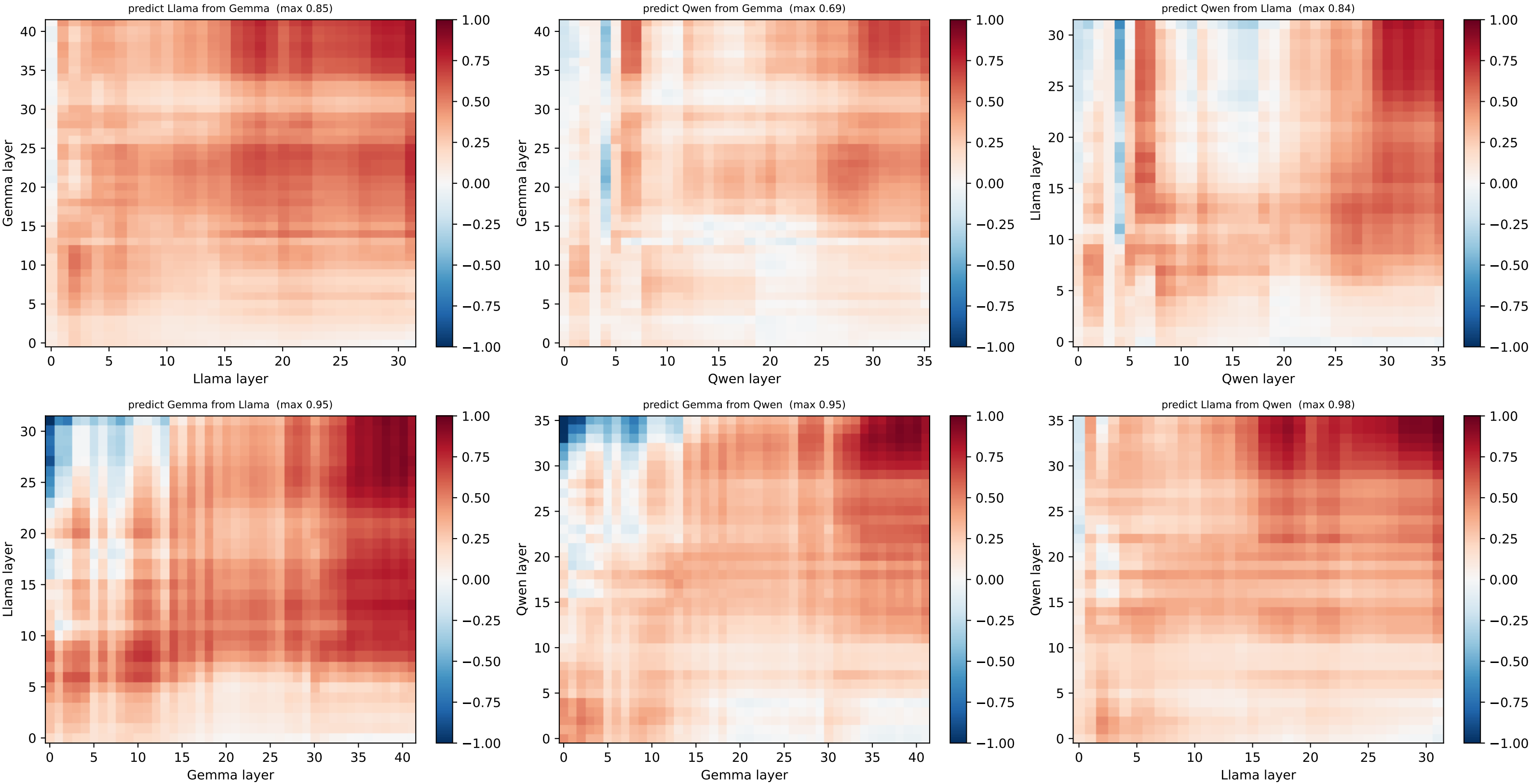
[grid] HELD-OUT node 3 @ (0.0, 3.0) — per-node R<sup>2</sup> layer×layer, both directions (red=well predicted, blue=worse than centroid)



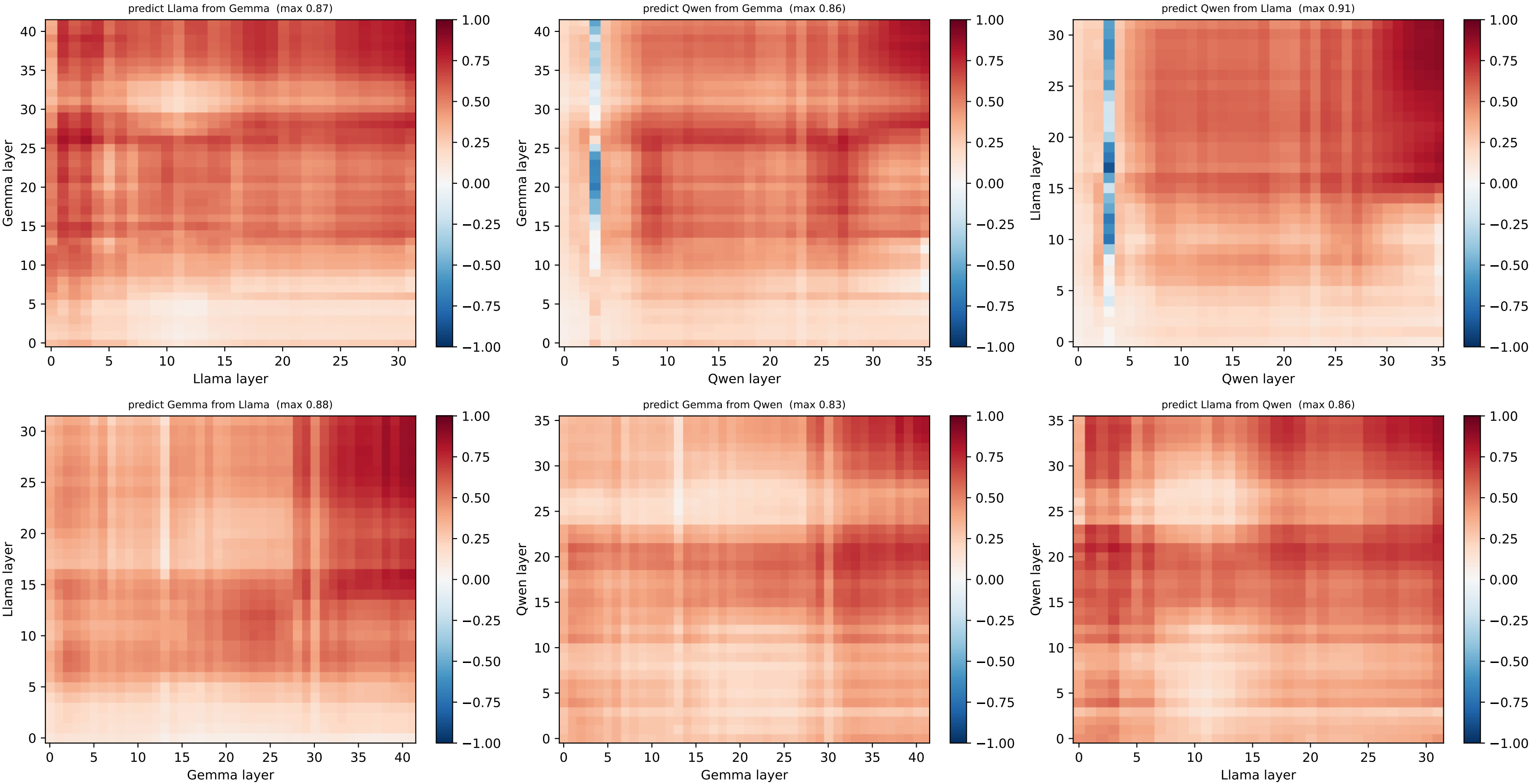
[grid] HELD-OUT node 4 @ (1.0, 0.0) — per-node R<sup>2</sup> layer×layer, both directions (red=well predicted, blue=worse than centroid)



[grid] HELD-OUT node 5 @ (1.0, 1.0) — per-node  $R^2$  layer×layer, both directions (red=well predicted, blue=worse than centroid)

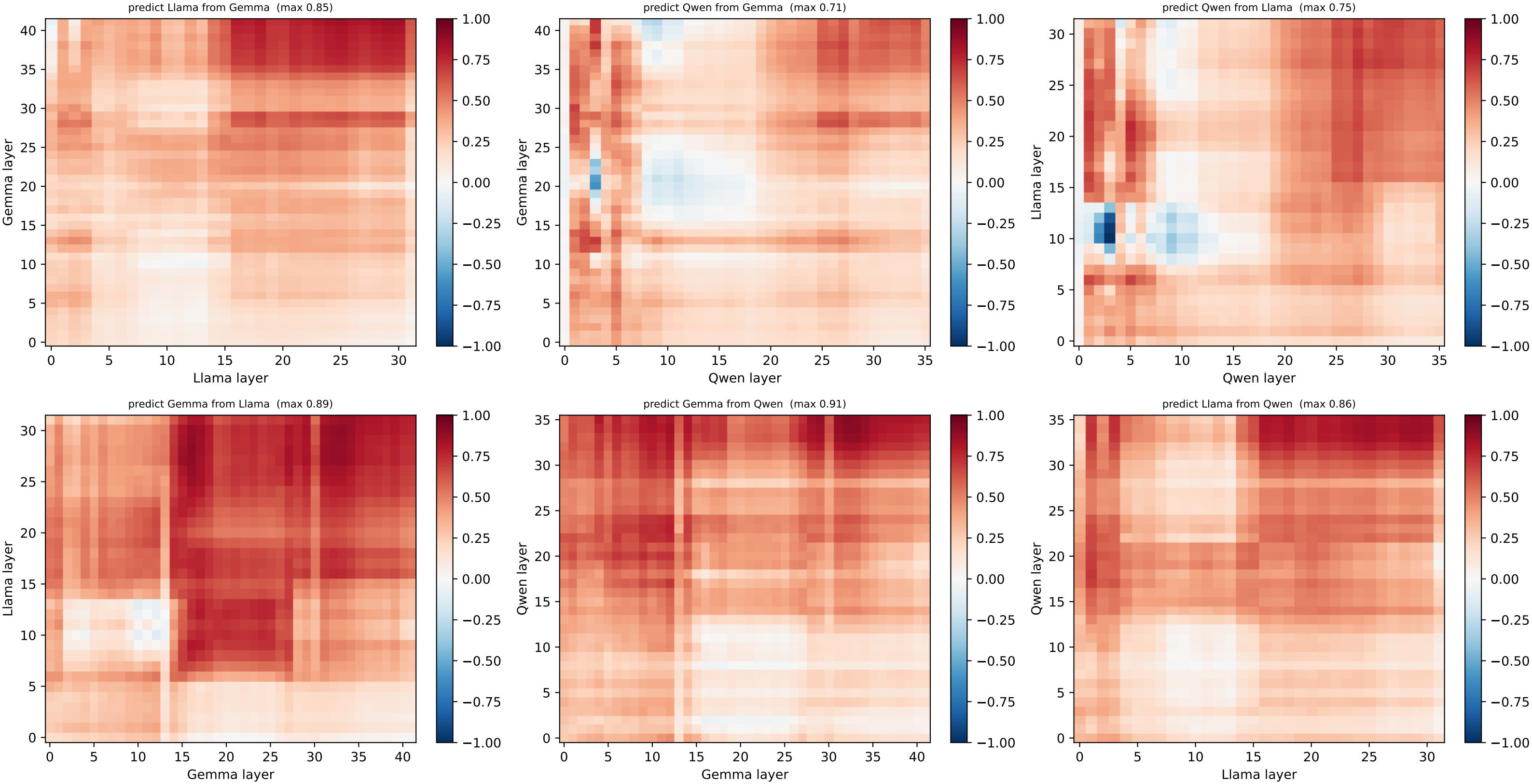


[grid] HELD-OUT node 6 @ (1.0, 2.0) — per-node R<sup>2</sup> layer×layer, both directions (red=well predicted, blue=worse than centroid)

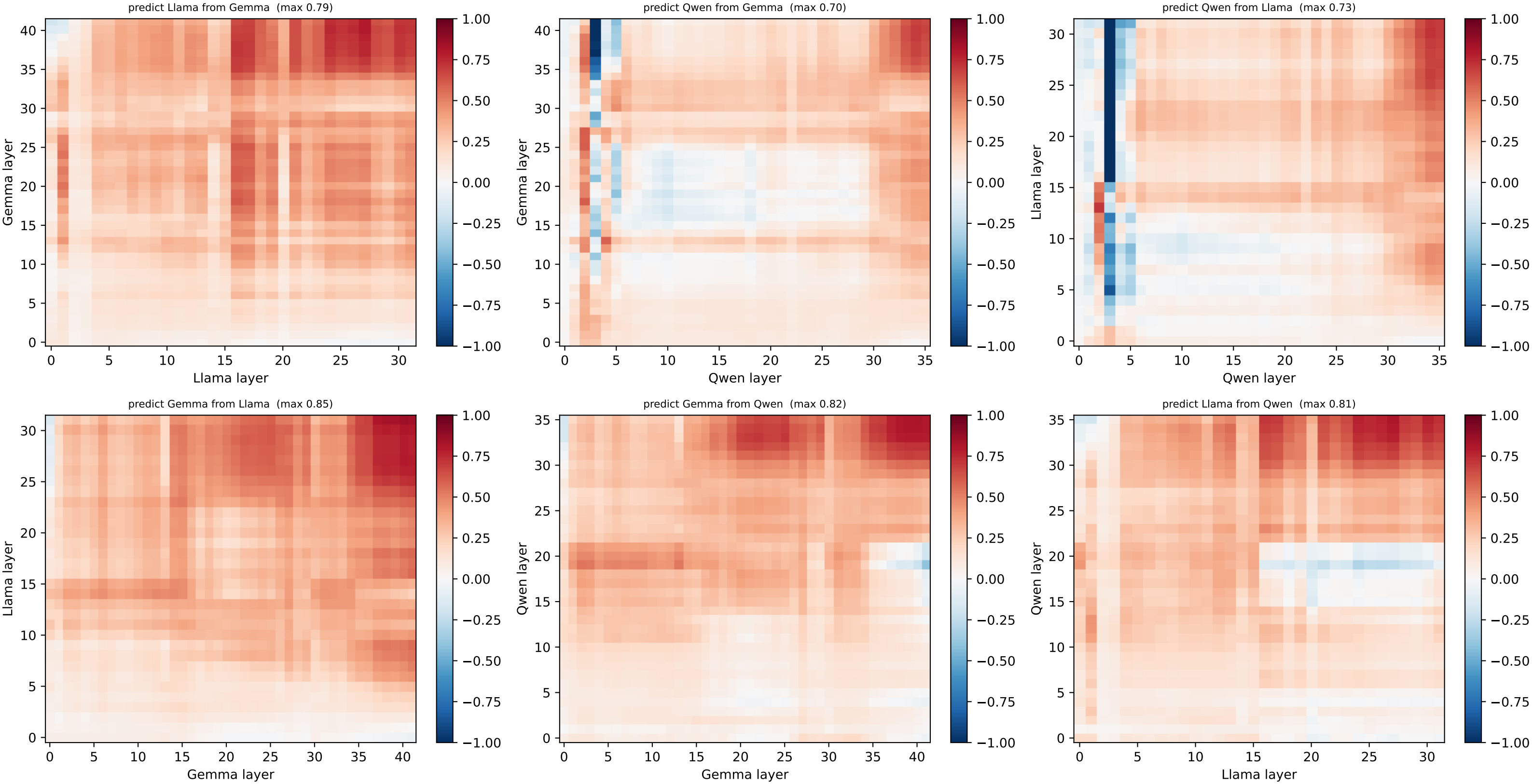




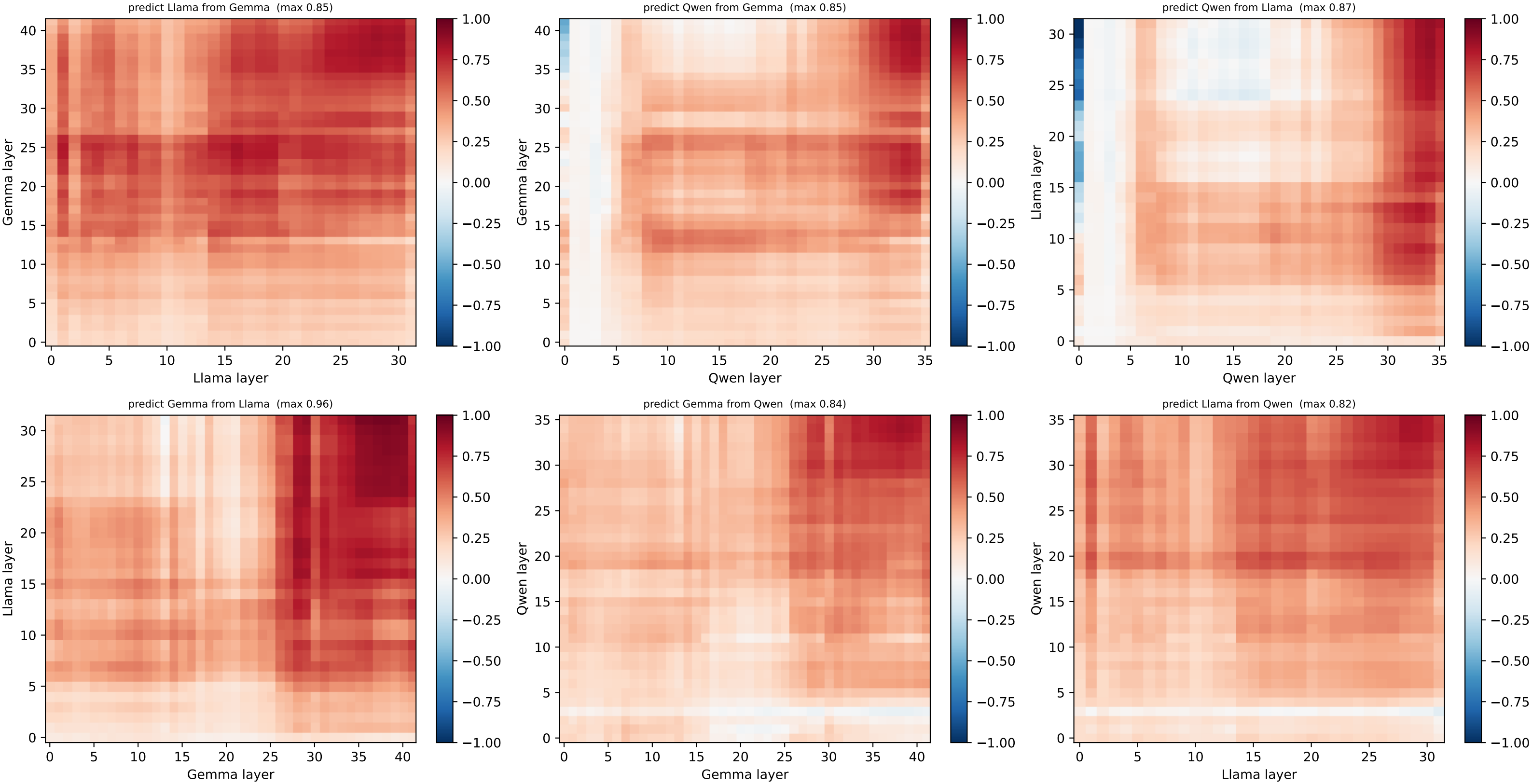
[grid] HELD-OUT node 7 @ (1.0, 3.0) — per-node  $R^2$  layer×layer, both directions (red=well predicted, blue=worse than centroid)



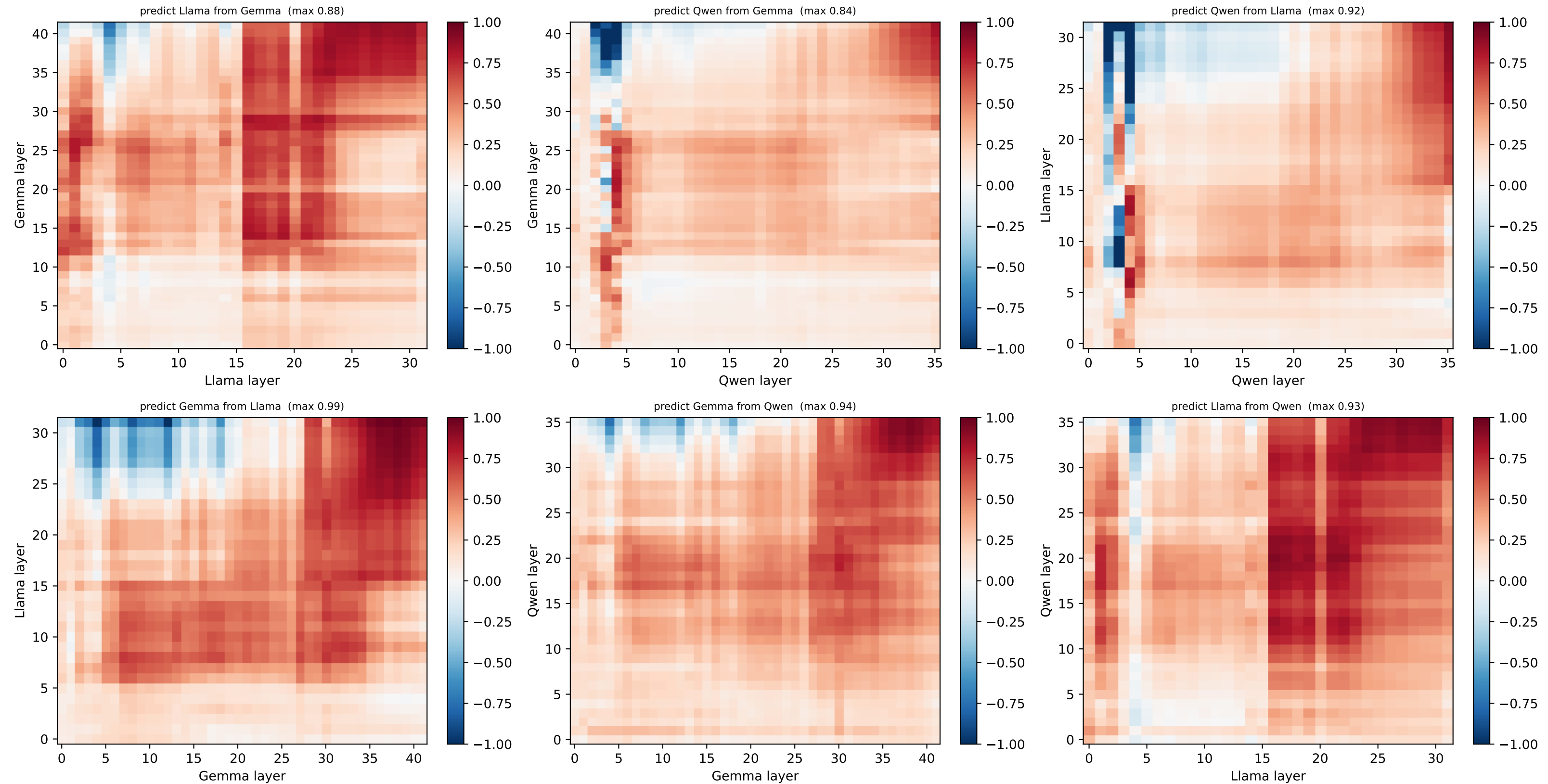
[grid] HELD-OUT node 8 @ (2.0, 0.0) — per-node R<sup>2</sup> layer×layer, both directions (red=well predicted, blue=worse than centroid)



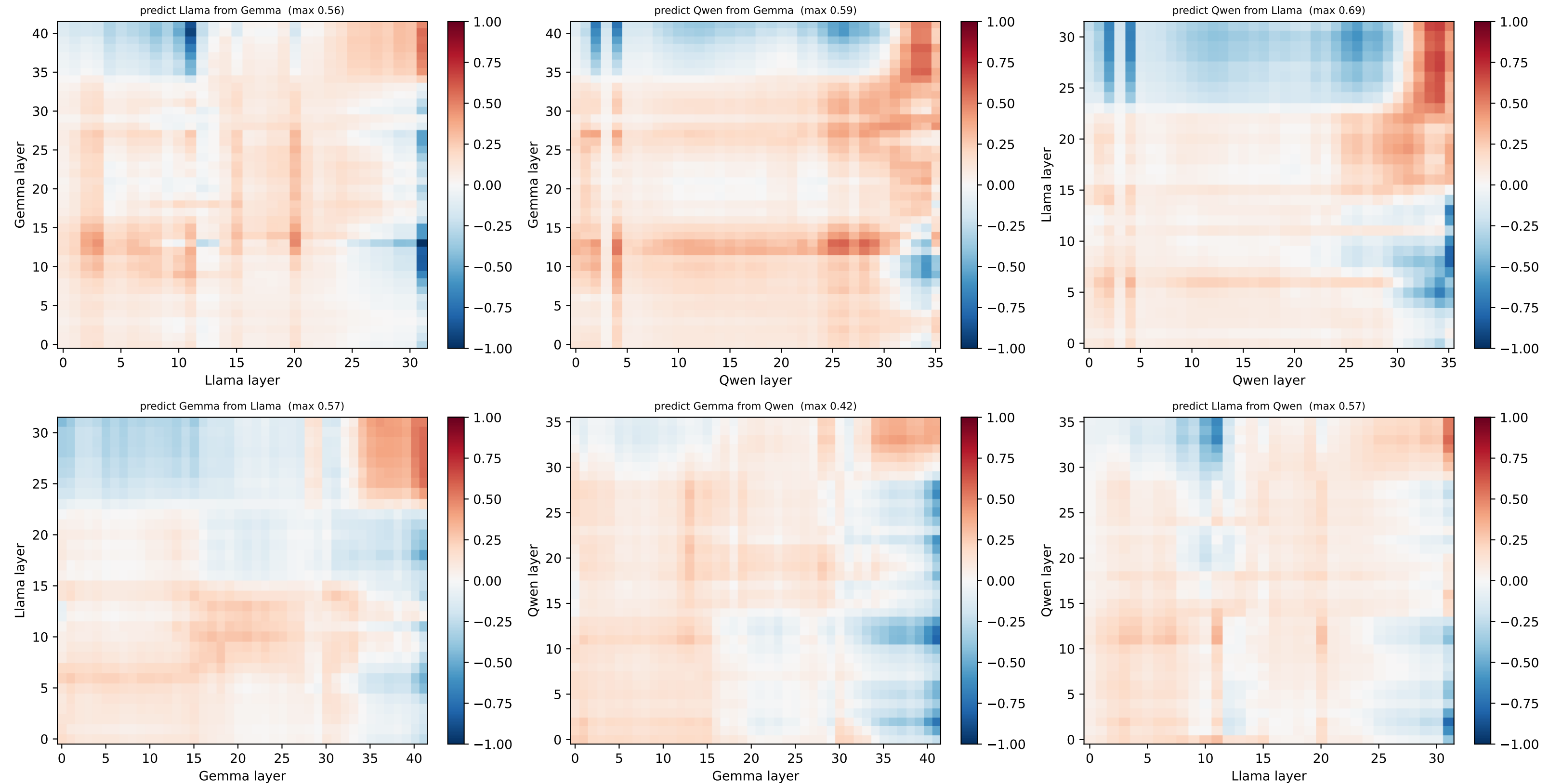
[grid] HELD-OUT node 9 @ (2.0, 1.0) — per-node  $R^2$  layer×layer, both directions (red=well predicted, blue=worse than centroid)



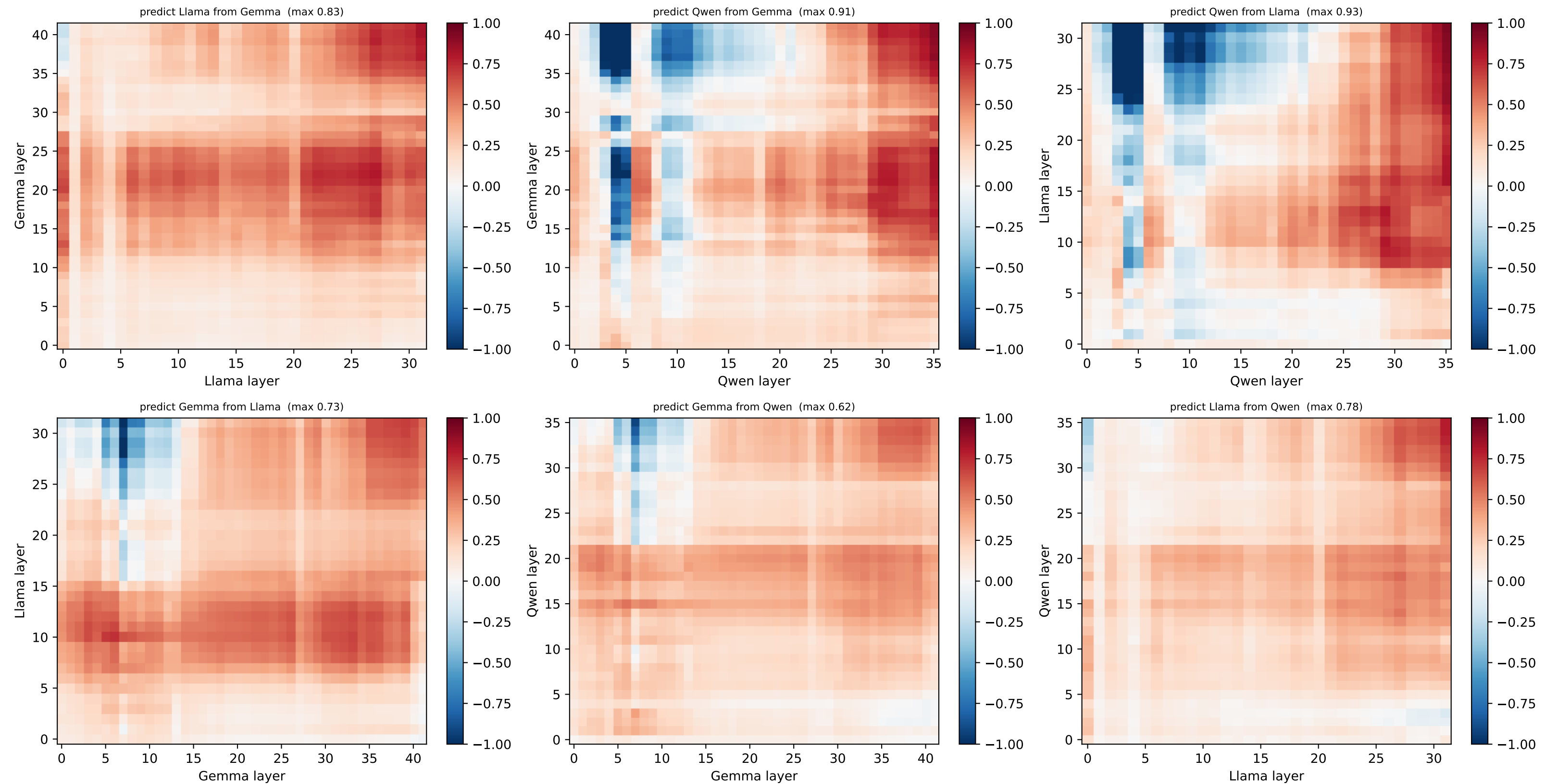
[grid] HELD-OUT node 10 @ (2.0, 2.0) — per-node  $R^2$  layer $\times$ layer, both directions (red=well predicted, blue=worse than centroid)



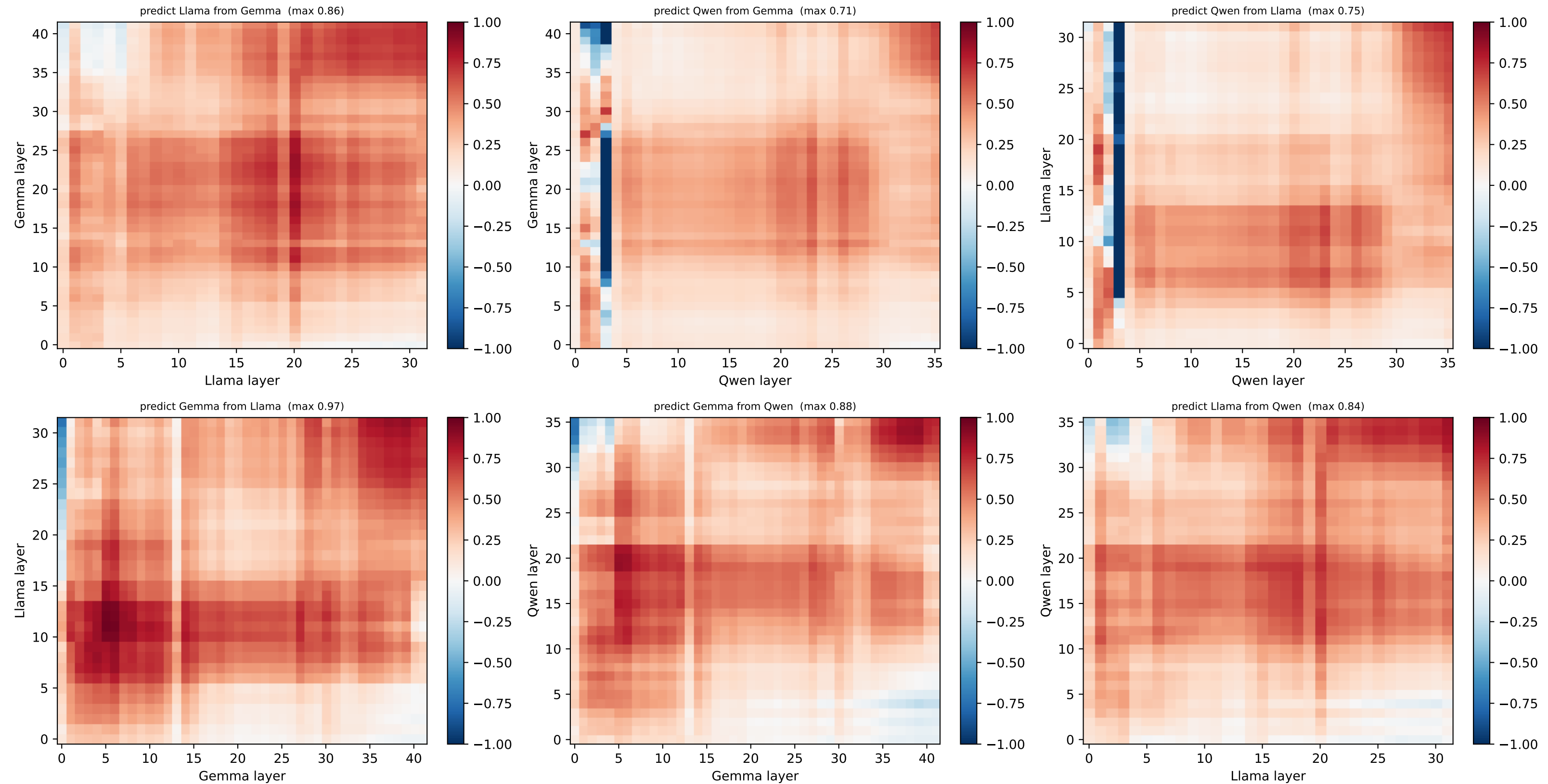
[grid] HELD-OUT node 11 @ (2.0, 3.0) — per-node  $R^2$  layer $\times$ layer, both directions (red=well predicted, blue=worse than centroid)



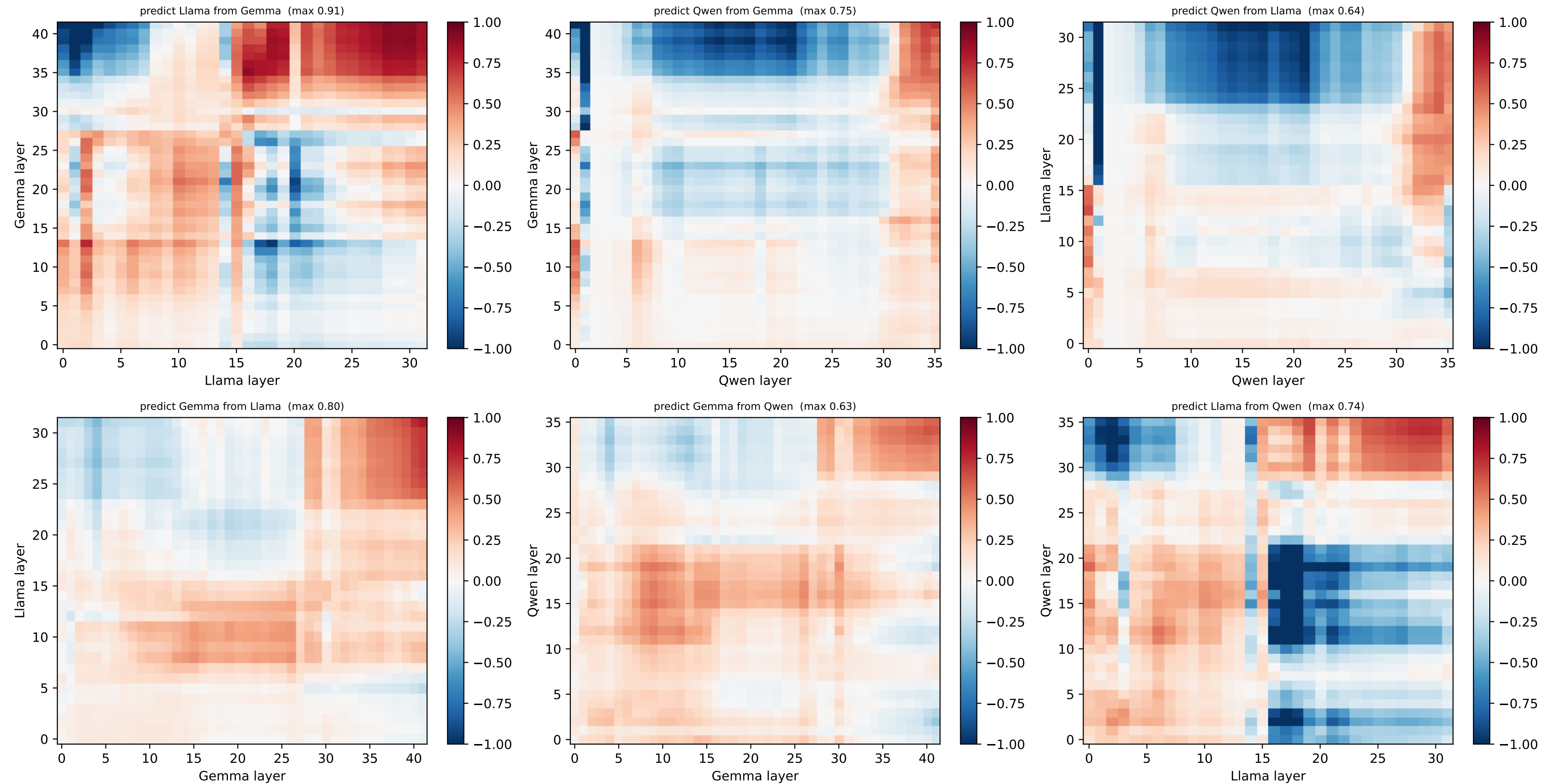
[grid] HELD-OUT node 12 @ (3.0, 0.0) — per-node  $R^2$  layer $\times$ layer, both directions (red=well predicted, blue=worse than centroid)



[grid] HELD-OUT node 13 @ (3.0, 1.0) — per-node  $R^2$  layer×layer, both directions (red=well predicted, blue=worse than centroid)



[grid] HELD-OUT node 14 @ (3.0, 2.0) — per-node  $R^2$  layer $\times$ layer, both directions (red=well predicted, blue=worse than centroid)





[grid] HELD-OUT node 15 @ (3.0, 3.0) — per-node  $R^2$  layer×layer, both directions (red=well predicted, blue=worse than centroid)

